Exercise 6.10 SOLUTION

Q1. Sum the following series up to n terms

i. $1\times3+2\times5+3\times7+\ldots$

Sol: First factor: $1,2,3,4,\ldots,n$ Here $a_1=1,d=1,T_k=a_1+(k-1)d=1+(k-1)1=1+k-1=k$

2nd Factor: $3,5,7,9,\ldots$ Here $a_1=3,d=2,T_k=a_1+(k-1)d=3+(k-1)2=3+2k-2=2k+1$

General term T_k is

$T_k=k\cdot(2k+1)$

Sum of the first n term is

$$S_n=\sum_{k=1}^{n}k(2k+1)=\sum_{k=1}^{n}(2k^2+k)$$

$$S_n=2\sum_{k=1}^{n}k^2+\sum_{k=1}^{n}k \qquad \sum_{k=1}^{n}k=\frac{n(n+1)}{2}$$

Now use the formulae

$$S_n=2\cdot\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}=\frac{n(n+1)}{6}\cdot[2(2n+1)+3]=\frac{n(n+1)}{6}\cdot(4n+5) \qquad \sum_{k=1}^{n}k^2=\frac{n(n+1)(2n+1)}{6}$$

$$S_n=\frac{n(n+1)(4n+5)}{6}$$

نوٹ

سوال کو دیکھیں تو ہر ٹرم میں دو نمبر ضرب کھا رہے ہیں تو ہر ٹرم کے پہلے نمبر سے T_k معلوم کریں پھر دوسرے نمبر سے T_k معلوم کریں پھر دونوں کو ضرب دیں تو S_n آجائے گا۔

ii. $1\times5+2\times8+3\times11+\ldots$

Sol: First factor: $1,2,3,4,\ldots,n$ Here $a=1,d=1,T_k=a_1+(k-1)d=1+(k-1).1=1+k-1=k$

2nd Factor: $5,8,11,\ldots$ Here $a=5,d=3,T_k=a_1+(k-1)d=5+(k-1)3=5+3k-3=3k+2$

General term T_k is $T_k=k\cdot(3k+2)$

Sum of the first n term is

$$S_n=\sum_{k=1}^{n}(3k^2+2k)=3\sum_{k=1}^{n}k^2+2\sum_{k=1}^{n}k \qquad \sum_{k=1}^{n}k=\frac{n(n+1)}{2}$$

Now by putting $\sum_{k=1}^{n}k^2=\frac{n(n+1)(2n+1)}{6}$

$$S_n=3\cdot\frac{n(n+1)(2n+1)}{6}+2\cdot\frac{n(n+1)}{2}=\frac{n(n+1)}{6}\cdot[3(2n+1)+6]$$

$$S_n=\frac{n(n+1)}{6}\cdot(6n+9)=\frac{n(n+1)(6n+9)}{6}=\frac{3n(n+1)(2n+3)}{6}=\frac{n(n+1)(2n+3)}{2}$$

iii. $1\times2+2\times5+3\times8+\ldots$

Sol: First factor: $1,2,3,4,\ldots,n$ Here $a_1=1,d=1,T_k=a_1+(k-1)d=1+(k-1).1=1+k-1=k$

2nd Factor: $2,5,8,\ldots$ here $a_1=2,d=3,T_k=a_1+(k-1)d=2+(k-1)3=2+3k-3=3k-1$

General term T_k is

$T_k=k\cdot(3k-1)$

Sum of the first n term is $S_n=\sum_{n=1}^{n}(3k^2-k)=3\sum_{k=1}^{n}k^2-\sum_{k=1}^{n}k$ $\qquad \sum_{k=1}^{n}k=\frac{n(n+1)}{2}$

Now by putting

$$S_n=3\cdot\frac{n(n+1)(2n+1)}{6}-\frac{n(n+1)}{2}=\frac{n(n+1)}{6}\cdot[3(2n+1)-3]=\frac{n(n+1)}{6}\cdot(6n) \qquad \sum_{k=1}^{n}k^2=\frac{n(n+1)(2n+1)}{6}$$

$$S_n=n(n+1)(n)=n^2(n+1)$$

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iv. $1\times3\times5+2\times4\times6+3\times5\times7+\ldots$

Sol: First Factor: $1,2,3,\ldots..n$ here $a_1=1,d=1,T_k=a_1+(k-1)d=1+(k-1)1=\cancel{1}+k-\cancel{1}=k$

2nd Factor: $3,4,5,\ldots..n$ Here $a_1=3,d=1,T_k=a_1+(k-1)d=3+(k-1)1=3+k-1=k+2$

3rd Factor: $5,6,7,\ldots..n$ Here $a_1=5,d=1,T_k=a_1+(k-1)d=5+(k-1)=5+k-1=k+4$

$$T_k=k\cdot(k+2)(k+4)$$

$$S_n=\sum_{k=1}^{n}k(k+2)(k+4)=k\left(k^2+6k+8\right)=k^3+6k^2+8k$$

$$S_n=\sum_{k=1}^{n}k\left(k^2+6k+8\right)$$

$$S_n=\sum_{k=1}^{n}k^3+6\sum_{k=1}^{n}k^2+8\sum_{k=1}^{n}k$$

Now by putting

$$S_n=\left[\frac{n(n+1)}{2}\right]^2+6\cdot\frac{n(n+1)(2n+1)}{6}+8\cdot\frac{n(n+1)}{2}$$

$$S_n=\frac{n^2(n+1)^2}{4}+n(n+1)(2n+1)+4n(n+1)=n(n+1)\left[\frac{n(n+1)}{4}+2n+1+4\right]=n(n+1)\left[\frac{n^2+n+8n+20}{4}\right]$$

$$=\frac{n(n+1)}{4}\left(n^2+9n+20\right)=\frac{n(n+1)}{4}\left(n^2+4n+5n+20\right)=\frac{n(n+1)}{4}\left[n(n+4)+5(n+4)\right]=\frac{n}{4}(n+1)(n+4)(n+5)$$

v. $1\times2\times4+2\times3\times7+3\times4\times10+\ldots$

Sol: 1st Factor: $1,2,3,\ldots.n$ Here $a_1=1,d=1,T_k=a_1+(k-1)d=1+(k-1)=1+k-\cancel{1}=k$

2nd Factor: $2,3,4,\ldots..,n$ Here $a_1=2,d=1,T_k=a_1+(k-1)d=2+(k-1)1=2+k-1=k+1$

3rd Factor: $4,7,10,\ldots..,n$ Here $a_1=4,d=3,T_k=a_1+(k-1)d=4+(k-1)3=4+3k-3=3k+1$

$$T_k=k\cdot(k+1)(3k+1)$$

$$T_k=k\left(3k^2+4k+1\right)=3k^3+4k^2+k$$

$$S_n=\sum_{k=1}^{n}\left(3k^3+4k^2+k\right)$$

$$S_n=3\sum_{k=1}^{n}k^3+4\sum_{k=1}^{n}k^2+\sum_{k=1}^{n}k$$

Now by putting

$$S_n=3\left[\frac{n(n+1)}{2}\right]^2+4\cdot\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}$$

$$=\frac{n(n+1)}{2}\left[\frac{3n(n+1)}{2}+\frac{4(2n+1)}{3}+1\right]$$

$$=\frac{n(n+1)}{2}\left[\frac{9n(n+1)+8(2n+1)+6}{6}\right]=\frac{n(n+1)}{2\times6}\left(9n^2+9n+16n+8+6\right)=\frac{n(n+1)}{2\times6}\left(9n^2+25n+14\right)$$

$$=\frac{n(n+1)}{12}\left(9n^2+25n+14\right)=\frac{n(n+1)\left(9n^2+18n+7n+14\right)}{12}=\frac{n(n+1)\left[9n(n+2)+7(n+2)\right]}{12}$$

$$=\frac{n(n+1)(9n+7)(n+2)}{12}$$

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vi. $2^2 + 4^2 + 6^2 + \ldots$

Sol: For $2,4,6,\ldots..n$ Here $a_1 = 2, d = 2$

$$T_k = \left[a_1 + (k-1)d\right]^2 = \left[2 + (k-1)2\right]^2 = \left[\cancel{2} + 2k - \cancel{2}\right]^2 = (2k)^2$$

$$T_k = (2k)^2 = 4k^2$$

$$S_n = \sum_{k=1}^{n} 4k^2 = 4\sum_{k=1}^{n} k^2$$

$$S_n = 4 \cdot \frac{n(n+1)(2n+1)}{6} \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$S_n = \frac{2n(n+1)(2n+1)}{3}$$

vii. $3^2 + 6^2 + 9^2 + \ldots$

Sol: For $3,6,9,\ldots..n$

Here $a_1 = 3, d = 3$

$$T_k = \left(a_1 + (k-1)d\right)^2 = \left(3 + (k-1)3\right)^2 = (3 + 3k - 3)^2$$

$$T_k = (3k)^2 = 9k^2$$

$$S_n = \sum_{k=1}^{n} 9k^2 = 9\sum_{k=1}^{n} k^2$$

$$S_n = 9 \cdot \frac{n(n+1)(2n+1)}{6} \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$S_n = \frac{3n(n+1)(2n+1)}{2}$$

viii. $4 \times 1^2 + 7 \times 2^2 + 10 \times 3^2 + \ldots$

Sol: 1st Factor $4,7,10,\ldots..n$

$a_1 = 4, d = 3, T_k = a_1 + (k-1)d = 4 + (k-1)3 = 4 + 3k - 3 = 3k + 1$

2nd Factor $1,2,3,\ldots..,n$

$$a_1 = 1, d = 1, T_k = \left[a_1 + (k-1)d\right]^2 = \left[1 + (k-1).1\right]^2 = \left[\cancel{1} + k - \cancel{1}\right]^2 = k^2$$

$$T_k = (3k+1)k^2 = 3k^3 + k^2$$

$$S_n = \sum_{k=1}^{n} \left(3k^3 + k^2\right) = 3\sum_{k=1}^{n} k^3 + \sum_{k=1}^{n} k^2$$

$$S_n = 3\left(\frac{n(n+1)}{2}\right)^2 + \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)}{2}\left[\frac{3n(n+1)}{2} + \frac{2n+1}{3}\right] = \frac{n(n+1)}{2}\left[\frac{9n(n+1) + 4n + 2}{6}\right]$$

$$= \frac{n(n+1)}{12}\left[9n^2 + 9n + 4n + 2\right] = \frac{n(n+1)\left(9n^2 + 13n + 2\right)}{12}$$

ix. $3 + (3+7) + (3+7+11) + \ldots$

Sol:

The series is:

$3 + (3+7) + (3+7+11) + \ldots$

Here $a_1 = 3$, $d = 4$, $n = r$ then

$a_r = 3 + (r-1)4 = 4r - 1$

The sum of the nth bracket is:

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$$S_n=\sum_{r=1}^{n}(4r-1)=4\sum_{r=1}^{n}r-\sum_{r=1}^{n}1=4\frac{n(n+1)}{2}-n=2n^2+2n-n=2n+n$$

The sum of the series up to n terms is:

$$S=\sum_{i=1}^{n}\left(2i^2+i\right)=2\sum_{i=1}^{n}i^2+\sum_{i=1}^{n}i$$

$$=2\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}=\frac{n(n+1)(2n+1)}{3}+\frac{n(n+1)}{2}=\frac{n(n+1)}{6}(4n+2+3)=\frac{n(n+1)(4n+5)}{6}$$

x. $\mathbf{1^2+\left(1^2+2^2\right)+\left(1^2+2^2+3^2\right)+\ldots}$

Sol: $$\sum_{k=1}^{n}\left(\sum_{j=1}^{k}j^2\right)$$

$$\sum_{k=1}^{n}j^2=\frac{k(k+1)(2k+1)}{6}$$

$$S_n=\sum_{k=1}^{n}\frac{k(k+1)(2k+1)}{6}$$

$$S_n=\frac{1}{6}\sum_{k=1}^{n}k(k+1)(2k+1)$$

$$S_n=\frac{1}{6}\sum_{k=1}^{n}\left(2k^3+3k^2+k\right)$$

$$S_n=\frac{1}{6}\left[2\sum_{k=1}^{n}k^3+3\sum_{k=1}^{n}k^2+\sum_{k=1}^{n}k\right]$$

$$S_n=\frac{1}{6}\left[2\left(\frac{n(n+1)}{2}\right)^2+3\cdot\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}\right]$$

$$=\frac{n(n+1)}{2\times6}\left[\frac{2n(n+1)}{2}+2n+1+1\right]=\frac{n(n+1)}{12}\left[n^2+n+2n+2\right]=\frac{n(n+1)}{12}\left(n^2+3n+2\right)$$

$$=\frac{n(n+1)}{12}\left(n^2+n+2n+2\right)=\frac{n(n+1)}{12}\left(n(n+1)+2(n+1)\right)$$

$$=\frac{n(n+1)(n+1)(n+2)}{12}=\frac{n(n+1)^2(n+2)}{12}$$

Q2. Sum the series

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i. $\mathbf{1^2-2^2+3^2-4^2+\ldots+(2n-1)^2-(2n)^2}$

Sol: $$T_n=(2n-1)^2-(2n)^2=\left(\cancel{4n^2}-4n+1-\cancel{4n^2}\right)=-4n+1$$

So $T_k=-4k+1$

$$S_n=\sum_{k=1}^{n}(-4k+1)=-4\sum_{k=1}^{n}k+\sum_{k=1}^{n}1$$

$$=-4\cdot\frac{n(n+1)}{2}+n=-2n(n+1)+n=-2n^2-2n+n=-2n^2-n$$

$$S_n=-2n^2-n=-n(2n+1)$$

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ii. $\frac{1^2}{1}+\frac{1^2+2^2}{2}+\frac{1^2+2^2+3^2}{3}+\ldots$ to n terms

Sol: Here $T_n=\frac{\frac{n(n+1)(2n+1)}{6}}{n}=\frac{1}{\cancel{n}}\left[\frac{\cancel{n}\left(2n^2+n+2n+1\right)}{6}\right]=\frac{1}{6}\left(2n^2+3n+1\right)$

So $T_k=\frac{1}{6}\left(2k^2+3k+1\right)$

$S_n=\sum_{k=1}^{n}T_k=\frac{1}{6}\sum_{k=1}^{n}\left(2k^2+3k+1\right)$

$=\frac{1}{6}\left[2\sum_{k=1}^{n}k^2+3\sum_{k=1}^{n}k+n\right]=\frac{1}{6}\left[\frac{2n(n+1)(2n+1)}{6}+3.\frac{n(n+1)}{2}+n\right]$

$=\frac{n}{6}\left[\frac{(n+1)(2n+1)}{3}+\frac{3(n+1)}{2}+1\right]=\frac{n}{6}\left[\frac{2\left(2n^2+3n+1\right)+9(n+1)+6}{6}\right]$

$=\frac{n}{36}\left(4n^2+6n+2+9n+9+6\right)=\frac{n}{36}\left(4n^2+15n+17\right)$

Q3. Find the sum of n terms of the series whose n^{th} terms are given

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i. $5n^2+2n+3$

Sol: $T_n=5n^2+2n+3$

$T_k=5k^2+2k+3$

$S_n=\sum_{k=1}^{n}T_k=\sum_{k=1}^{n}\left(5k^2+2k+3\right)$

$S_n=5\sum_{k=1}^{n}k^2+2\sum_{k=1}^{n}k+\sum_{k=1}^{n}3=5.\frac{n(n+1)(2n+1)}{6}+2.\frac{n(n+1)}{2}+3n$

$S_n=\frac{5n(n+1)(2n+1)}{6}+n(n+1)+3n=n\left[\frac{5(n+1)(2n+1)}{6}+n+1+3\right]=n\left[\frac{5(n+1)(2n+1)}{6}+n+4\right]$

$=n\left[\frac{5\left(2n^2+3n+1\right)+6n+24}{6}\right]=\frac{n}{6}\left[10n^2+15n+5+6n+24\right]=\frac{n}{6}\left(10n^2+21n+29\right)$

ii. n^2+2n-3

Sol: $T_n=n^2+2n-3$ so $T_k=k^2+2k-3$

$S_n=\sum_{k=1}^{n}\left(k^2+2k-3\right)$

$S_n=\sum_{k=1}^{n}k^2+2\sum_{k=1}^{n}k-\sum_{k=1}^{n}3=\frac{n(n+1)(2n+1)}{6}+\cancel{2}\frac{n(n+1)}{\cancel{2}}-3n$

$=\frac{n(n+1)(2n+1)}{6}+n(n+1)-3n=n\left(\frac{(n+1)(2n+1)}{6}+n+1-3\right)=n\left(\frac{2n^2+3n+1+6n-12}{6}\right)$

$S_n=\frac{n}{6}\left(2n^2+9n-11\right)$

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Q4. Given n^{th} term of the series, find the sum to $2n$ terms

i. $3n^2+5n+2$

Sol: $T_n = 3n^2+5n+2 \quad T_k = 3k^2+5k+2$

نوٹ

T_n دیا گیا ہے پہلے S_n معلوم کریں پھر n کی جگہ 2n لگا کر S_{2n} معلوم کریں۔

$$S_n = \sum_{k=1}^{n}\left(3k^2+5k+2\right)$$

$$S_n = 3\sum_{k=1}^{n}k^2+5\sum_{k=1}^{n}k+\sum_{k=1}^{n}2 = 3\cdot\frac{n(n+1)(2n+1)}{6}+5\cdot\frac{n(n+1)}{2}+2n$$

$$S_n = \frac{3n(n+1)(2n+1)}{6}+\frac{5n(n+1)}{2}+2n = n\left[\frac{(n+1)(2n+1)}{2}+\frac{5(n+1)}{2}+2\right] = n\left[\frac{2n^2+3n+1}{2}+\frac{5n+5}{2}+2\right]$$

$$= \frac{n}{2}\left[2n^2+3n+1+5n+5+4\right] = \frac{n}{2}\left(2n^2+8n+10\right) = n\left(n^2+4n+5\right)$$

Replace n by $2n$

$$S_{2n} = 2n\left[(2n)^2+4(2n)+5\right] = 2n\left(4n^2+8n+5\right)$$

ii. n^2+n-2

Sol: $T_n = n^2+n-2 \quad T_k = k^2+k-2$

$$S_n = \sum_{k=1}^{n}\left(n^2+n-2\right)$$

$$S_n = \sum_{k=1}^{n}k^2+\sum_{k=1}^{n}k-\sum_{k=1}^{n}2 = \frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}-2n$$

$$= n\left(\frac{(n+1)(2n+1)}{6}+\frac{n+1}{2}-2\right) = n\left(\frac{2n^2+3n+1}{6}+\frac{n+1}{2}-2\right) = n\left(\frac{2n^2+3n+1+3n+3-12}{6}\right) = \frac{n}{6}\left(2n^2+6n-8\right)$$

$$S_n = \frac{n}{3}\left(n^2+3n-4\right)$$

Replace n by $2n$

$$S_{2n} = \frac{2n}{3}\left((2n)^2+3(2n)-4\right)$$

$$S_{2n} = \frac{2n}{3}\left(4n^2+6n-4\right) = \frac{4n\left(2n^2+3n-2\right)}{3}$$

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